

Massimo Piazza

Aerospace Systems Eng. | Computer Vision / AI

@ massimopiazza97@gmail.com

☎ (+39) 3297788437

in linkedin.com/in/mpiazza97

🔗 squareospace.com

📅 Date of Birth: July 14, 1997

🇮🇹 Citizenship: Italy

📍 Berlin (Germany)



Working Experience

Mission Integration Engineer

Exolaunch 🔗 exolaunch.com

📅 Apr 2026 – Present

📍 Berlin, DE



System Budgets Engineer

Rivada Space Networks 🔗 rivadaspace.com

📅 Nov 2023 – Oct 2025

📍 Berlin, DE



- Systems Engineer responsible for system-wide technical budgets of a 600-satellite LEO constellation; the network of laser-meshed satellites is designed to provide Carrier Ethernet services via regenerative Ka-band phased-array payloads
- Requirements/interface management and cross-segment design integration, across Space/Ground/User segments + V&V support
- Toolchain integration between E2E simulation, MBSE modeling, reqs. management, and V&V, via automated workflows and SW interfaces that improved traceability and sped up design

Lead Spacecraft Navigation Engineer

Infinite Orbits 🔗 infiniteorbits.io

📅 Apr 2022 – Nov 2023

📍 Toulouse, FR



- Leading development & integration of onboard software of a proximity navigation camera / star tracker
- Systems engineering & Agile PM: AIV/AIT support, reqs. management, procurement, documentation
- Deep learning-based pose estimation (CNNs) & Kalman filtering
- Synthetic image generation (Blender/UE5 + flight dynamics sim.)
- Development of software for star tracker online calibration and performance characterization
- Integration & testing of GEO flight dynamics software
- Project coordinator for a R&D contract awarded by CNES
- 7-DoF robotic testbed integration, calibration, operations
- Business development support for proposal writing & tracking of US federal grant/contract opportunities
- Other: product management, recruiting, building optical testbeds

Computer Vision Engineer

Infinite Orbits

📅 Jan 2021 – Mar 2022

📍 Toulouse, FR

- Development of an image processing pipeline (Matlab/Simulink) for angles-only navigation, enabling detection and tracking of resident space objects
- Auto-coding & HIL testing of image processing and estimation algorithms (Xilinx target hardware + FreeRTOS)
- Development of orbit determination software (Orekit-based)
- Other: link budgets, high-fidelity SRP & power generation modeling
- Founding engineer (#6)

Education

MSc in Space Engineering

Politecnico di Milano

📅 Sep 2018 – Dec 2020



Thesis: "Deep learning-based monocular relative pose estimation of uncooperative spacecraft"
(English taught)

BSc in Aerospace Engineering

Politecnico di Milano

📅 Oct 2015 – Sep 2018

Final project: "Falcon 9 2nd stage redesign using LOX/LH₂ bipropellant"

Publications

Journal papers

- M. Piazza *et al.*, "Monocular relative pose estimation pipeline for uncooperative resident space objects," *AIAA (JAIS)*, pp. 1–20, 2022.

Conference proceedings

- D. Kaidanovic, M. Piazza, *et al.*, "Deep learning based relative navigation about uncooperative space objects," in *73rd International Astronautical Congress*, 2022.
- F. Ventre, ..., M. Piazza, *et al.*, "Phase-A design of a Mars south pole exploration mission: MARS PENGUIN," in *73rd International Astronautical Congress*, 2022.
- M. Piazza *et al.*, "Deep learning-based monocular relative pose estimation of uncooperative spacecraft," in *8th European Conference on Space Debris, ESA/ESOC*, 2021.

Peer-review activity

- Peer-reviewer for scientific journals, including: *Advances in Space Research (Elsevier)* and *Journal of Aerospace Information Systems (AIAA)*

Early Working Experience

Co-founder & CTO

WatchApp, Inc.

Jan 2015 – Aug 2019

Lewes, DE, USA



- Wrote the entire source-code of the mobile apps released by the startup (Canvas Keyboard, Everboard)
- Development of handwriting recognition algorithms and integration of several REST APIs

Project Experience

Telemetry & Telecommand Engineer

MARs South Pole Exploration and Geysers in-situ Investigation (MARS-PENGUIN)

Mar 2020 – Jul 2020

Milan, IT

- Phase-A design of a rover mission to the south pole of Mars, aimed at collecting soil samples and performing in-situ investigation of geyser phenomena
- TMTC design and link budget analysis involving: cruise, EDL and Martian surface operations

Head of Aerodynamics and Flight Dynamics

Skyward Experimental Rocketry skywarder.eu

Oct 2017 – Jul 2018 (Head)

Milan, IT

Oct 2016 – Sep 2017 (Dept. member)



- Carried out flight dynamics simulations and CFD analyses, aimed at designing a jet-powered UAV (project Cyrano)
- Interviewed ~50 people and selected new team members

Skills & Background

Advanced

- Astrodynamics
- Propulsion Systems
- AOCS
- Telecommunications
- Systems Engineering
- Structural Dynamics
- Machine Learning
- Computer Vision
- Orbit Determination
- Attitude Determination

- Matlab
- Simulink
- LaTeX
- Git
- Unix
- Python
- PyTorch
- C
- Objective-C
- Auto-coding
- Orekit
- Valispace
- Xcode
- CEA
- Adobe Illustrator/Photoshop/Premiere Pro
- MS Office
- Jira/Confluence

Intermediate

- GNC
- Flight Dynamics
- Aerodynamics
- Remote Sensing
- Signal Processing
- Optics
- LLMs
- OpenCV
- Blender
- SolidWorks
- Solid Edge
- Autodesk Inventor
- Vitis (Xilinx)
- Capella MBSE
- HTML/CSS

Patents

WO 2023094347 (issued Nov 24, 2023)
Vision-based Autonomous Navigation System and Method for a Satellite

Accomplishments

- Pose Estimation Challenge, ESA (2020)
- “Best Freshmen of the A.Y. 2015/16” – Politecnico di Milano (Mar 2017)
- Apple's 20 Under 20 (Jun 2015)
- Smau App Awards – Rome (2014)
- Smau App Awards – Milan (2013)

Certifications

- “Agile Systems Engineering” by UC Boulder
- “Deep Learning Specialization (5 courses)” by deeplearning.ai
- “Machine Learning” by Stanford University
- “Generative AI with LLMs” by AWS
- “Private Equity & Venture Capital” by Bocconi University
- “Agile Project Management” by Google
- “Finance for Managers” by IESE Business School
- GRE:⁽²⁰¹⁷⁾ 309/340 (Q:159 - V:150)

Languages

- English: full professional proficiency
- Italian: mother tongue
- French: limited working proficiency
- German: elementary proficiency

Other Interests

Music education

- Studied at Conservatory “A. Scarlatti”
Classical Guitar (8 years)
Musical Composition (4 years)
- Winner of multiple national and international classical music competitions

Various

- Entrepreneurship
- Venture Capital
- Financial markets
- Geopolitics
- Artificial Intelligence
- NewSpace